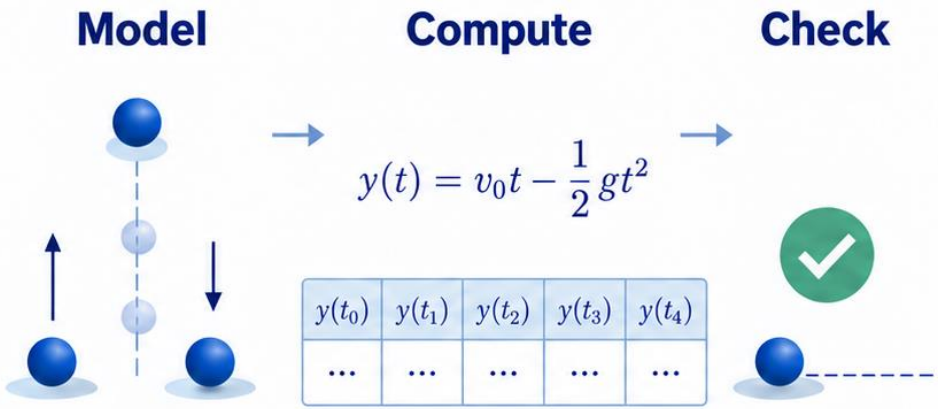
Computational Methods in Physics

PHY4605

Dr. Muhammad Khairul Adib

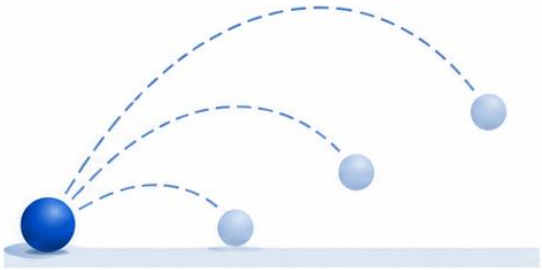
Turn a physical question into a computation you can explain and trust.



Why compute in physics?

Motion

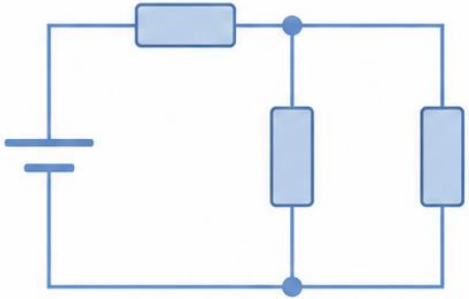
Change launch speed



Explore a parameter change

Circuit

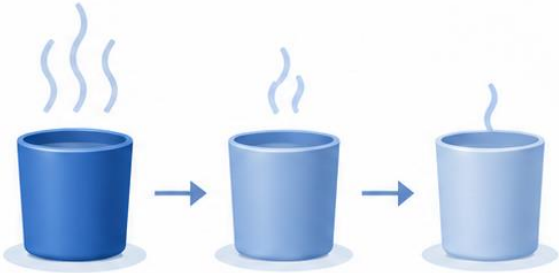
Solve coupled currents



Approximate a difficult solution

Cooling

Follow temperature over time

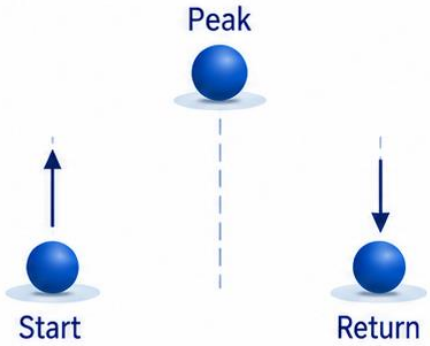


Investigate how a system behaves

A result needs a physical check

Predict

Vertical motion



Sketch the motion before coding.

Compute

$$y(t) = v_0 t - \frac{1}{2} g t^2$$

Upward positive; no air resistance

Height samples

$y(t_0)$	$y(t_1)$	$y(t_2)$	$y(t_3)$	$y(t_4)$
…	…	…	…	…

Use the model to calculate height.

Check

$y(0) = 0$ m

Starts at the launch height

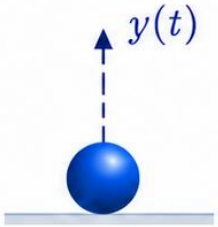


Check a known value, then interpret.

Explain the model. Check the result. Trust the evidence.

What you will learn to do

1 Identify the model and units

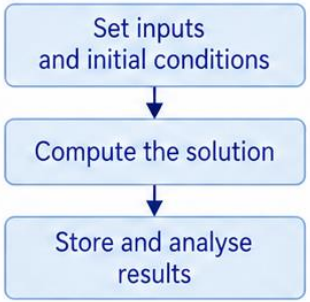


$y(t)$: vertical position (m)

$v(t)$: vertical velocity $(\mathrm{m\,s^{-1}})$

g : gravitational acceleration $(\mathrm{m\,s^{-2}})$

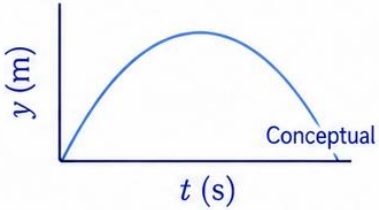
2 Plan an algorithm



3 Read, complete or modify code

```
1  % set parameters
2  % compute solution
3  % plot and analyse results
```

4 Validate and explain the result



Result is physically reasonable.

Explain what the result means in physical terms.

Start with supplied scaffolds. Build confidence through practice.

How we build a trustworthy computation

Define

Physical model • Scale and units

1 Physical model



Identify the system and idealisations

2 Scale and units

$[\mathrm{L}]$ $[\mathrm{M}]$ $[\mathrm{T}]$

Choose variables and consistent units

Implement

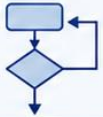
Discretisation • Algorithm • Code

3 Discretisation



Replace the continuous problem with a discrete model

4 Algorithm



Design a clear step-by-step procedure

5 Code



Implement the algorithm in code

Evaluate

Error and uncertainty • Validation • Physical interpretation

6 Error and uncertainty



Estimate numerical error and uncertainty

7 Validation



Check against known results or limiting cases

8 Physical interpretation



Explain the results in physical terms

Refine and improve

Your journey: build the foundations

1 Represent

Week 1

Arrays and plots

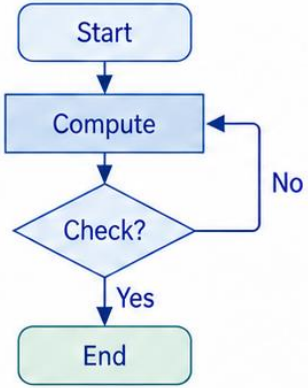


Vertical motion

2 Trace

Week 2

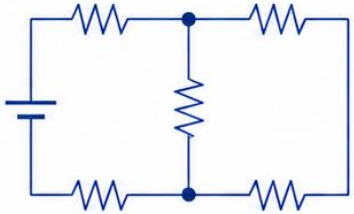
Algorithms and debugging



3 Solve

Week 3

Linear systems

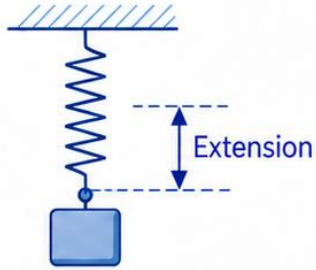


Two-loop circuit

4 Compare

Week 4

Parameter sweeps



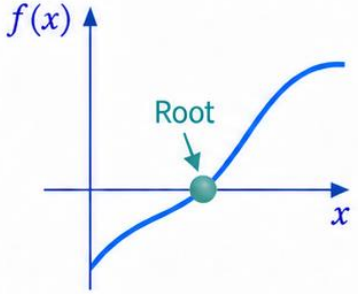
Spring extension

Familiar physics. Supplied code. One check at a time.

Your journey: learn numerical methods

1 Locate

Week 5
Root finding

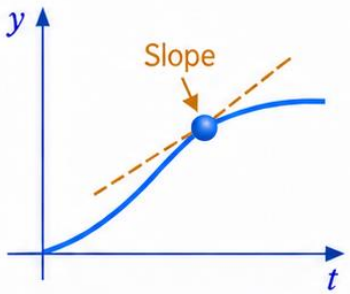


Conceptual

A physical solution

2 Estimate

Week 6
Numerical differentiation

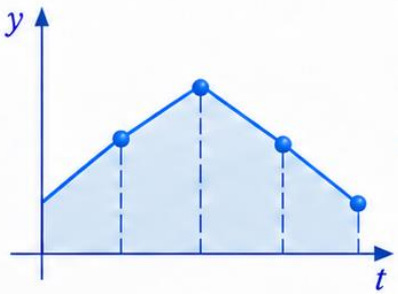


Conceptual

A physical rate

3 Accumulate

Week 7
Numerical integration

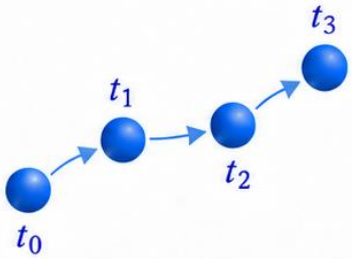


Conceptual

An accumulated quantity

4 Evolve

Week 8
ODE simulation



A changing state

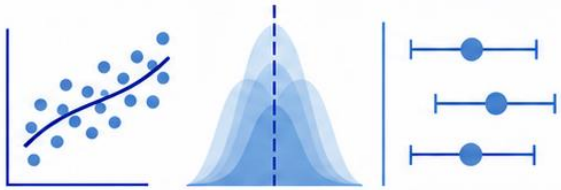
Known reference. Step-size check. Explain the result.

Your journey: investigate and defend

1 Explore

Weeks 9–11

Data and uncertainty

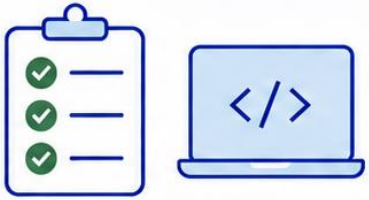


Data fitting (9) • Monte Carlo (10)
Sensitivity analysis (11)

2 Audit

Week 12

Method selection



Capstone studio

3 Defend

Weeks 13–14

Computational investigation



Present and explain
Reproduce and hand off

Week 14 continues this phase; no new core topic.

Present. Defend. Reproduce and hand off.

Use each course material for a purpose



Lecture slides (.pptx)
Follow the main ideas.



Learning notes (.pdf)
Revisit explanations and examples.



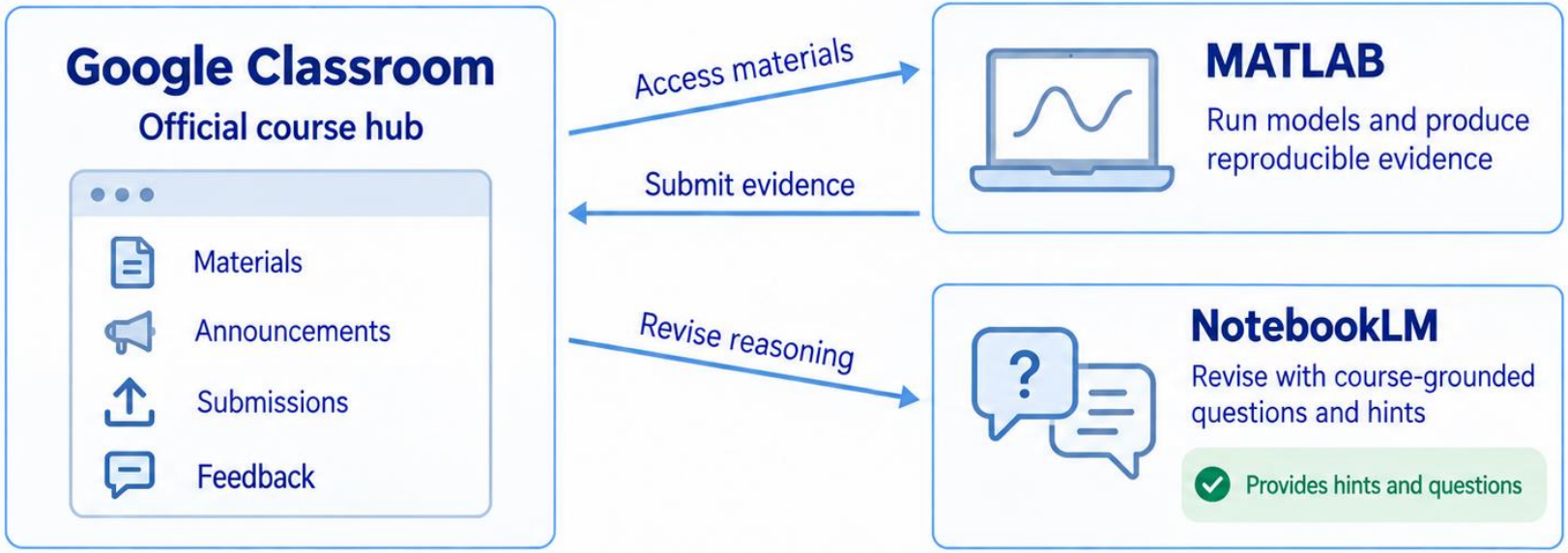
MATLAB Live Scripts (.m)
Run, modify and check the model.



Practical challenges
Investigate and record evidence.

Start with the materials in the relevant week.

Where learning and submission happen



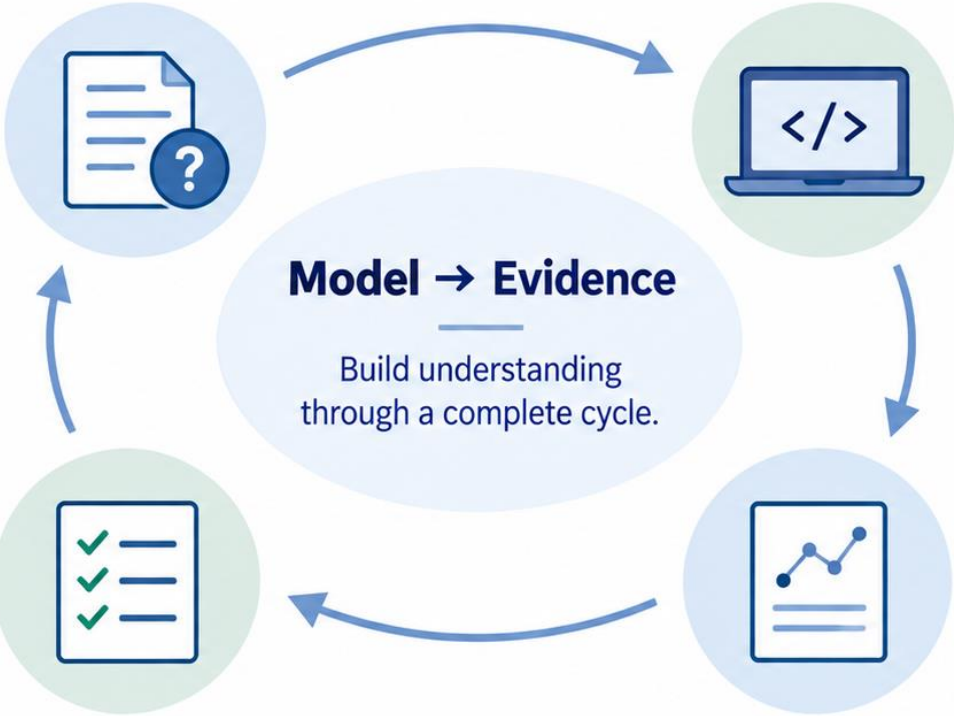
Your weekly learning routine

Before

Read, predict and complete the diagnostic when released.

During

Trace, modify and check.



After

Submit, explain and reflect.

Practical

Investigate and record evidence.

Rerun from a fresh MATLAB session.

How your learning is assessed

Written assessments

Open book • AI-free

Test 1	10%
Test 2	10%
Final examination	40%

Computational investigations

AI allowed and declared

Practical portfolio	15%
Individual lab test	10%
Capstone investigation	15%

Show the model, reasoning, checks and physical meaning.

Use AI and take responsibility for the result.

1 Investigate



Groups of 2–3;
ten micro-challenges.

2 Verify



Check independently;
rerun from a fresh session.

3 Submit



Record AI decisions and
lock the evidence.

4 Explain

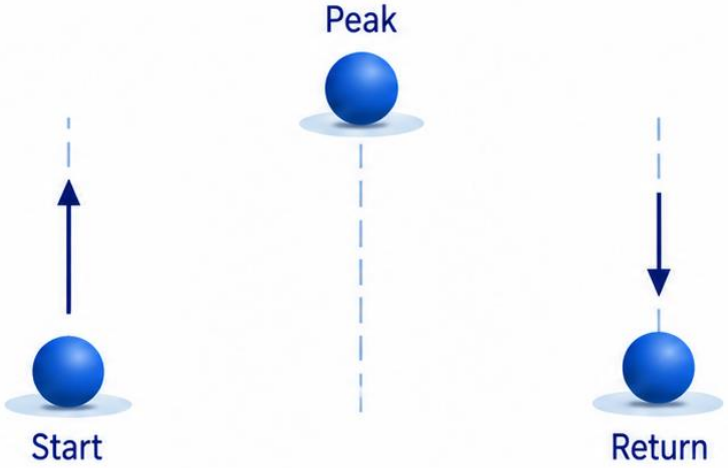


A question and one eligible
member are drawn after
submission.

Every member must understand the model, code and result.

Start with a prediction

- Complete MATLAB Onramp
- Check access to Google Classroom and MATLAB
- Open the Week 1 materials



What should the height–time graph of an upward-thrown ball look like?

Upward positive; no air resistance.